



Get your hands dirty with CloudNativePG

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KCD Czech & Slovak,
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<3 OSS

<3 Coding

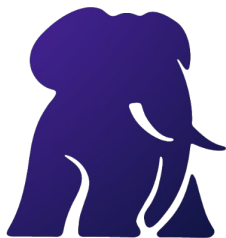
🎵 Playing sax

<3 Paola and Maria

Survey

CloudNativePG

- **CloudNativePG** is a CNCF Sandbox project. It is an open-source operator designed to manage **PostgreSQL** workloads on any supported **Kubernetes** cluster
- It defines a new K8s resource called **Cluster**, representing a PostgreSQL cluster made up of a single primary and an optional number of replicas



CloudNativePG



PostgreSQL

PostgreSQL

PostgreSQL is an object-relational **database** management system.

The first PostgreSQL release formed version 6.0 on January 1997.

Latest stable release is 18.4 (May 2026).

Because of the **liberal license**, PostgreSQL can be used, modified, and distributed by anyone free of charge for any purpose, be it private, commercial, or academic.

Full official documentation: postgresql.org/docs/current

PostgreSQL - Features

It supports a large part of the SQL standard and offers many modern features:

- Complex queries
- Foreign keys
- Triggers
- Updatable views
- Transactional integrity
- Multiversion Concurrency Control

PostgreSQL - Extensibility

PostgreSQL can be extended by the user in many ways, for example by adding new:

- Data types
- Functions
- Operators
- Aggregate functions
- Index methods
- Procedural languages

PostgreSQL - Love

PostgreSQL continues to outclass legacy and specialty databases in the most critical contexts - including technical performance, flexibility and applicability across the broadest number of enterprise workloads.

Open source is at the heart of this.

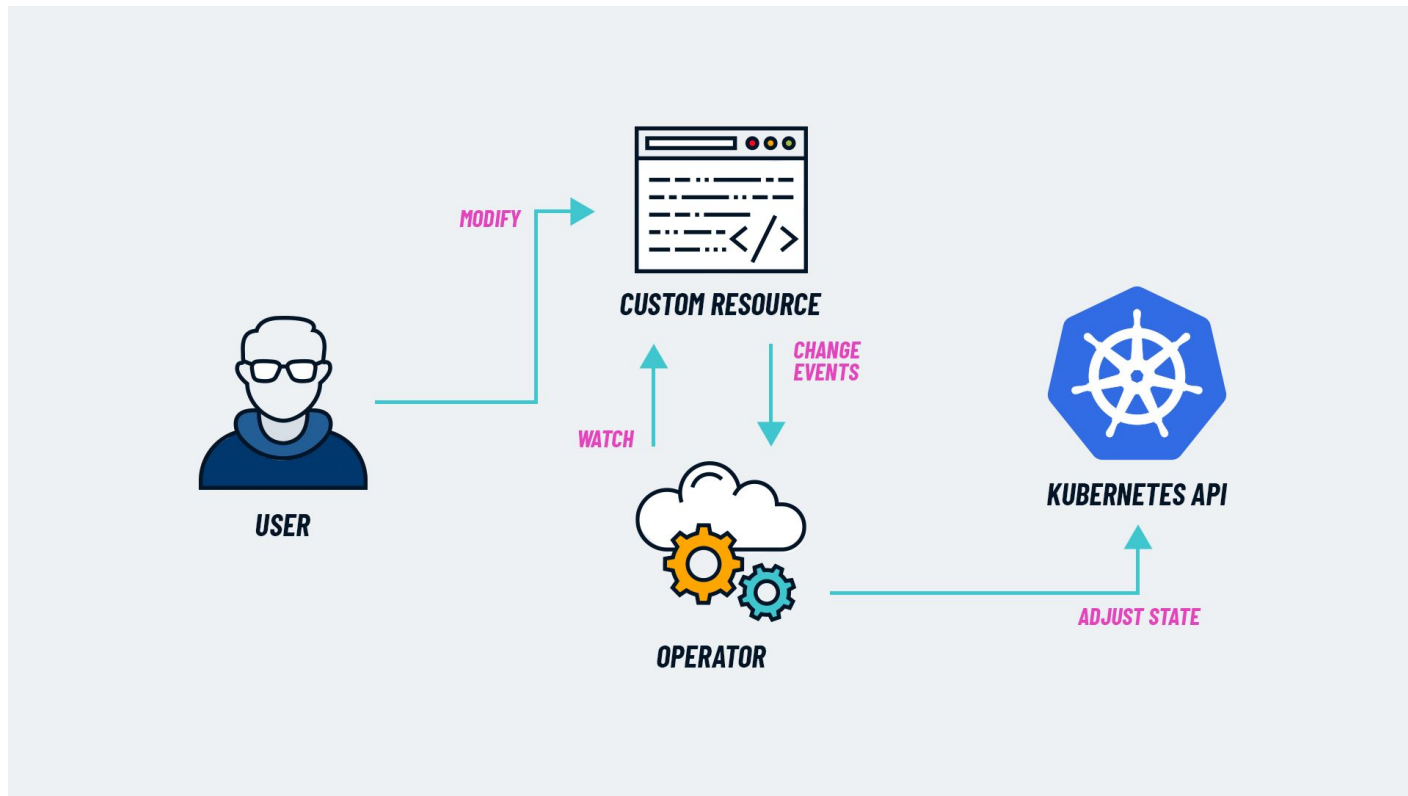
Developers **<3** Postgres.

Most popular Database: Stack Overflow Survey 2025, 2024 and 2023 .



Kubernetes operators

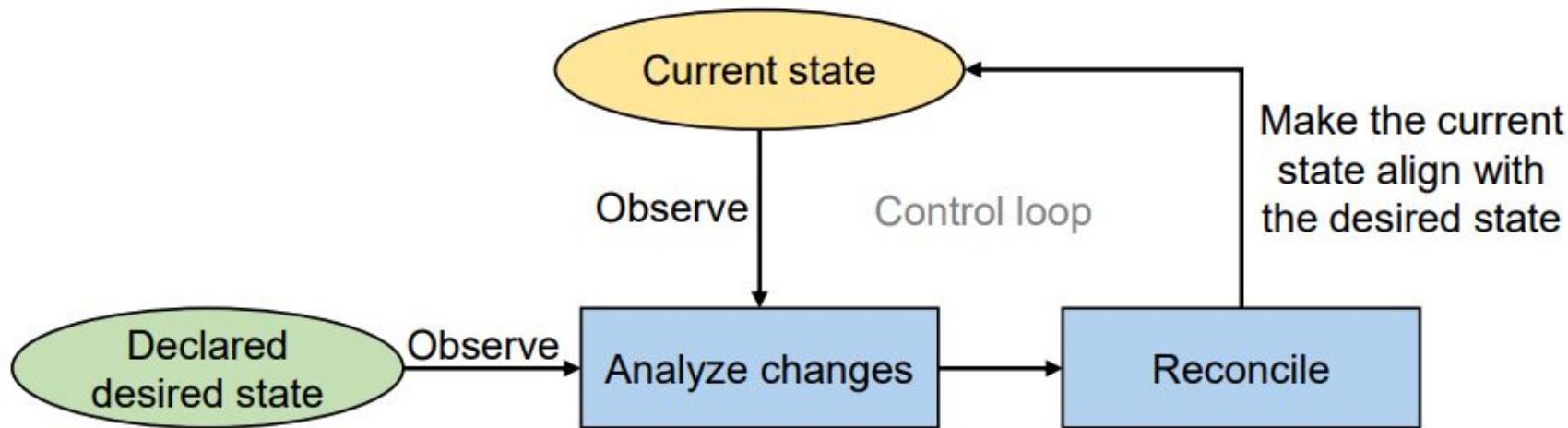
Kubernetes Operator



Kubernetes Operator - In a nutshell

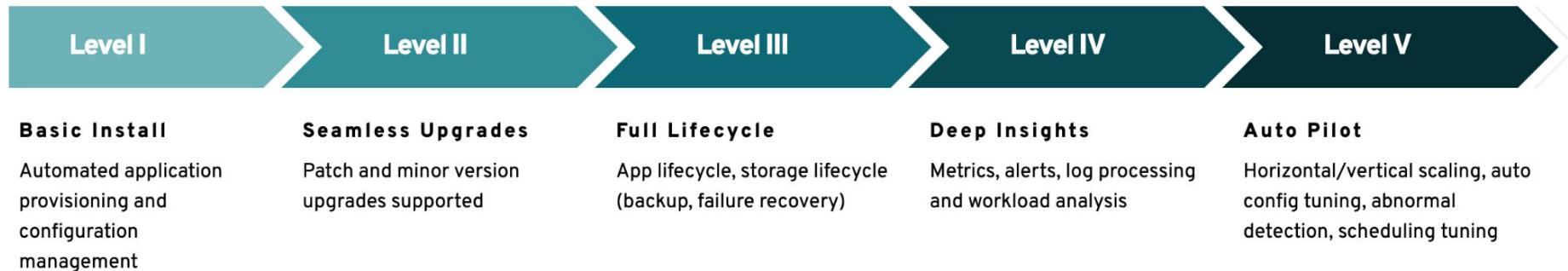
- Software extensions to Kubernetes
- Use of **custom resources** to manage applications and their components
- Operators follow Kubernetes principles, notably the control loop
- Operators implement and automate common Day-1 (installation, configuration, etc) and Day-2 (re-configuration, update, backup, failover, restore, etc.) activities
- Via operators, you can stop treating an application as a collection of primitives like Pods, Deployments, Services or ConfigMaps, but instead as a single object that only exposes the knobs that make sense for the application

Kubernetes Operator - Control loop



Picture from "[An Empirical Study on Kubernetes Operator Bugs](#)" by Gu - Xiao -Wei

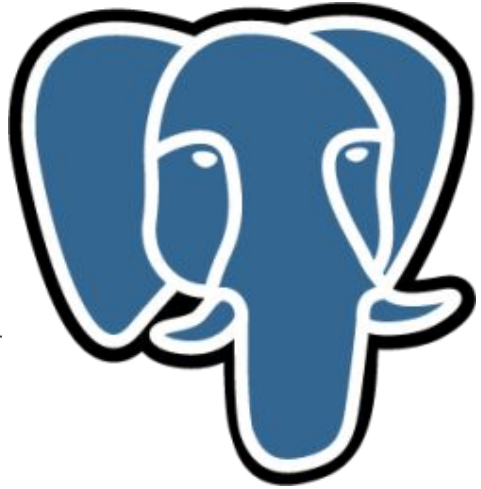
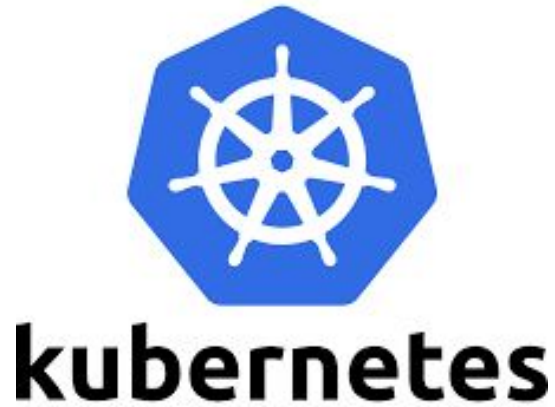
Kubernetes Operator - Capability levels



Kubernetes Operator - Write your own one

- **SDK** for the major languages
- Custom kubernetes **controllers**
- Design declarative **APIs** for operators
- Usually interact/control a custom object (Custom Resource Definitions (**CRD**))

Bring Kubernetes to Postgres, bring Postgres to Kubernetes!



Deep into CloudNativePG

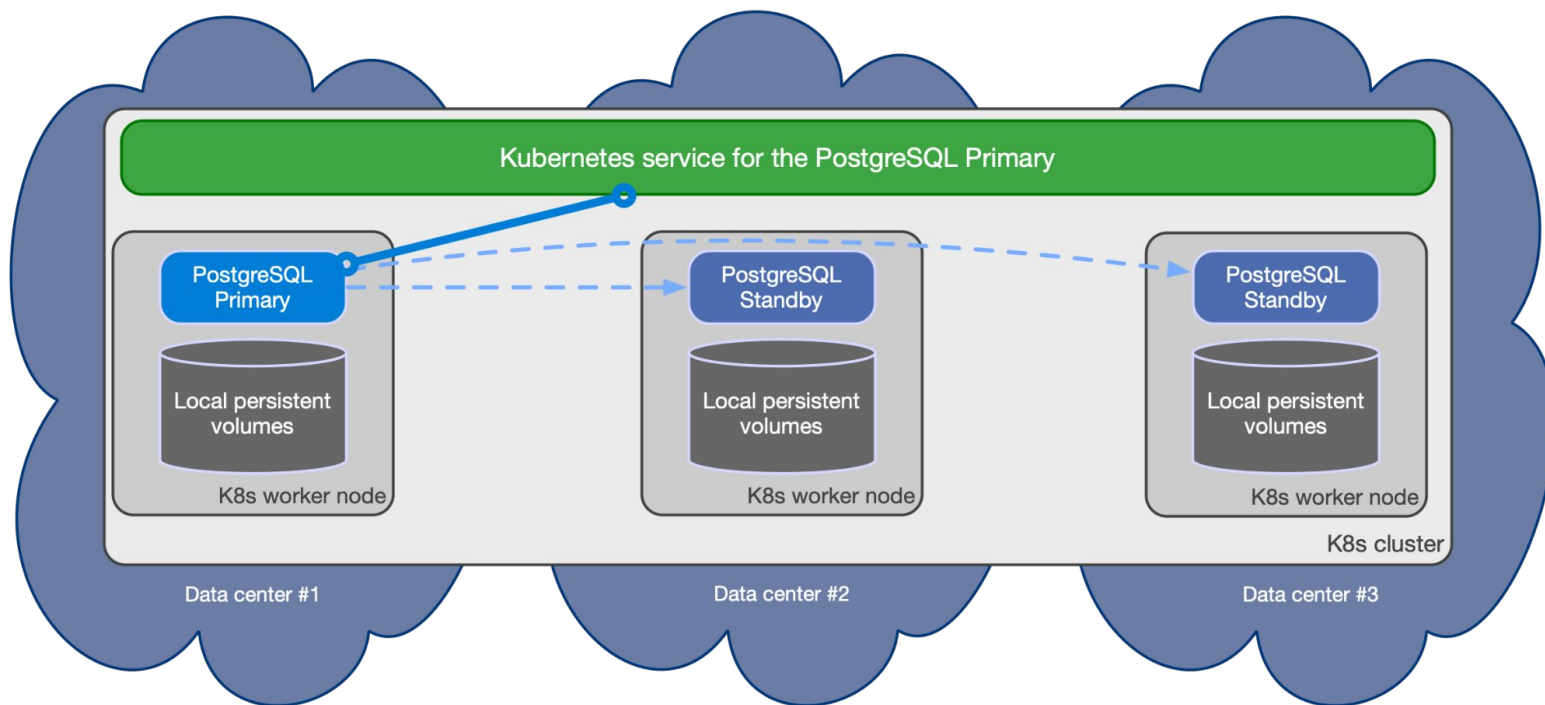
CloudNativePG - Cluster CRD example

```
> batcat cloudnative-pg/docs/src/samples/cluster-example.yaml
```

```
File: cloudnative-pg/docs/src/samples/cluster-example.yaml
```

```
1  apiVersion: postgresql.cnpg.io/v1
2  kind: Cluster
3  metadata:
4    name: cluster-example
5  spec:
6    instances: 3
7
8    storage:
9      size: 1Gi
```

CloudNativePG - Suggested architecture



CloudNativePG - PVC management

- **No Statefulsets**
- **The operator manages directly PersistentVolumeClaims:**
 - It creates a PVC for each PostgreSQL instance, with the goal of storing the PGDATA
 - It then mounts it into each pod
- **CSI** for PVCs allows
 - **Volume Snapshot** backup and restore
 - Online resize
 - PVC cloning

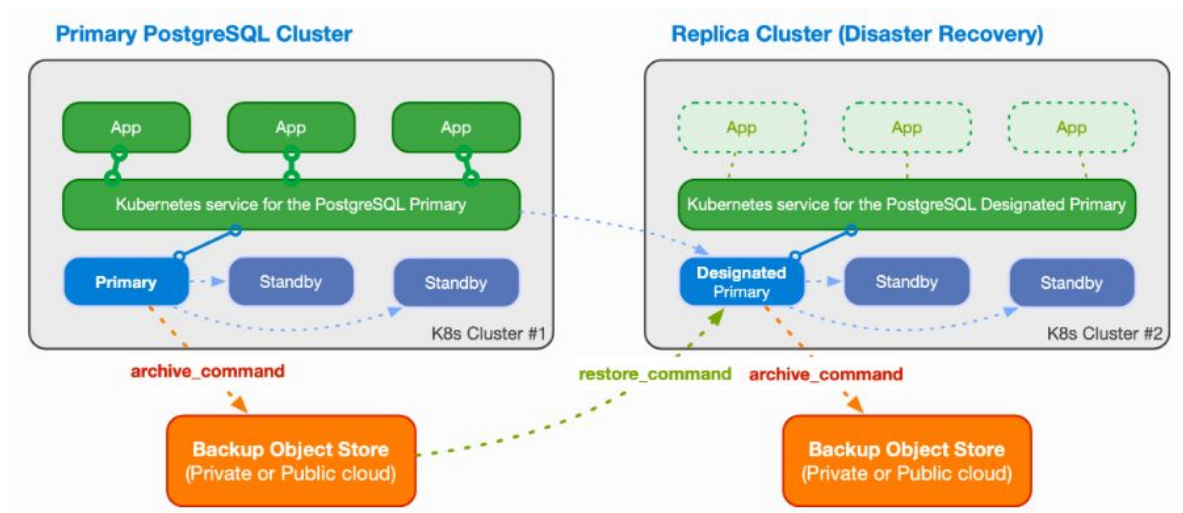
CloudNativePG - Main features

- Direct integration with K8s Api Server for self healing
- High Availability capability
- Planned switchover
- Scale up/down capabilities
- Definition of an arbitrary number of instances, rw and ro services
- Declarative management of PostgreSQL configuration
- Support for local PV with PVC templates
- Backup on VolumeSnapshot and object stores
- Major PG upgrade
- kubectl plugin for cluster management
- Declarative tablespaces, databases, publications and subscriptions
- CNPG-I: define your own plugin!

CloudNativePG - Deployment across Kubernetes clusters

Distributed PostgreSQL topology with **replica clusters** enables multi-cluster deployments in private, public, hybrid, and multi-cloud contexts.

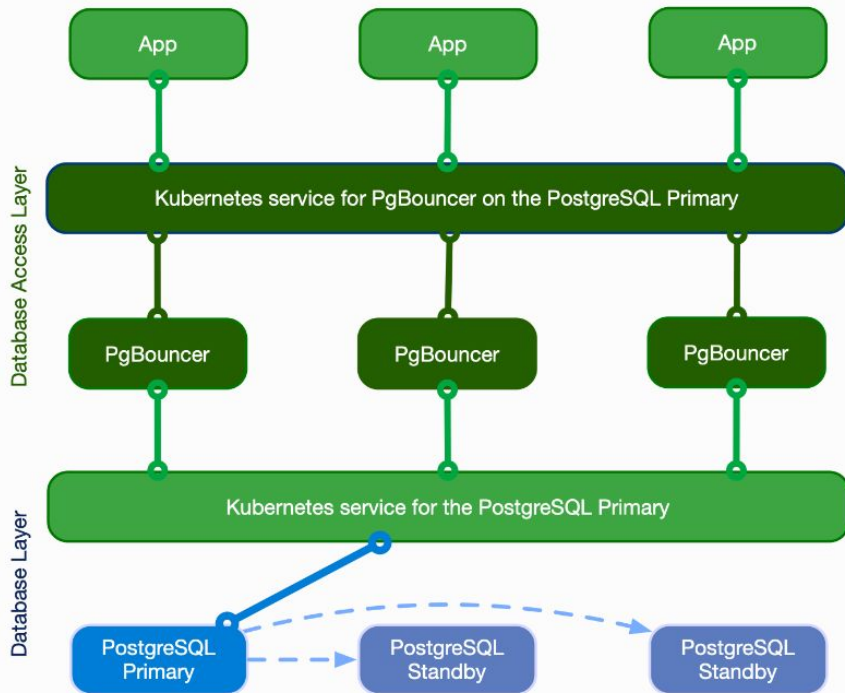
A replica cluster is a separate **Cluster** resource that is in continuous recovery, replicating from another source, either via WAL or streaming replication.



CloudNativePG - Connection pooling

CloudNativePG provides native support for connection pooling with PGBouncer, through the **Pooler** CRD.

A pooler is a deployment of PgBouncer pods that sits between your application and a PostgreSQL service. It creates a separate, scalable, configurable, and highly available database access layer.



CloudNativePG - Installation



```
kubectl apply --server-side -f \  
https://raw.githubusercontent.com/cloudnative-pg/cloudnative-pg/main/releases/cnpg-1.24.1.yaml
```

Refer to the CloudNativePG documentation for Helm Chart and Operator Lifecycle Manager (OLM), directly from OperatorHub.io

CloudNativePG - Operand images

- The PostgreSQL operand **container images** are available for all PGDG supported versions of PostgreSQL, across multiple architectures, directly from the postgres-containers project's GitHub Container Registry.
- Daily jobs ensure that CVEs in the entire stack are promptly addressed
- Additionally, the community provides images for the PostGIS extension
- Check this blog article, if you want create your own **custom** container image.

CloudNativePG - Observability

- CloudnativePG exposes native **Prometheus** exporters. Some are already defined, but you can extend them and add new ones
- All logs are in stdout and in **JSON** format
- CNPG plugin **status** command



CloudNativePG - Kubectl Plugin

CloudnativePG provides a **plugin** for kubectl to manage a cluster in Kubernetes

```
> ./kubectl-cnpg --help
A plugin to manage your CloudNativePG clusters

Usage:
  kubectl cnpg [command]

Operator-level administration
  install      CloudNativePG installation-related commands

Troubleshooting
  logs         Logging utilities
  report       Report on the operator or a cluster for troubleshooting

Cluster administration
  destroy      Destroy the instance named [cluster]-[node] or [node] with the associated PVC
  fencing      Fencing related commands
  hibernate    Hibernate related commands
  maintenance  Sets or removes maintenance mode from clusters
  promote      Promote the pod named [cluster]-[node] or [node] to primary
  reload       Reload a cluster
  restart      Restart a cluster or a single instance in a cluster

Database administration
  backup       Request an on-demand backup for a PostgreSQL Cluster
  certificate   Create a client certificate to connect to PostgreSQL using TLS and Certificate authentication
  publication   Logical publication management commands
  snapshot     DEPRECATED (use 'backup -m volumeSnapshot' instead)
  status       Get the status of a PostgreSQL cluster
  subscription Logical subscription management commands

Miscellaneous
  fio          Creates a fio deployment, pvc and configmap
  pgadmin4     Creates a pgAdmin deployment
  pgbench      Creates a pgbench job
  psql         Start a psql session targeting a CloudNativePG cluster

Additional Commands:
  completion   Generate the autocompletion script for the specified shell
  help         Help about any command
  version      Prints version, commit sha and date of the build
```

CloudNativePG - What I personally love

- Massive suite of e2e running in CI/CD pipelines
- Exhaustive documentation
- CNPG team members
- Community enthusiasm

The picture shows some of the maintainers running a demo at the Project Pavillion of KubeCon Eu London 2025



CloudNativePG - Product

- CloudNativePG is **production ready!**
- CloudNativePG became the de facto standard to deploy **PostgreSQL in Kubernetes**
- Community is very active inside the Cloud Native Computing Foundation channels: **cloudnativepg-dev** and **cloudnativepg-users**
- CloudNativePG has been accepted into the **CNCF Sandbox**

Health score

● Healthy

The Insights Health Score combines the four key areas to measure an open source project's overall trustworthiness. [Learn more](#)

Contributors

Popularity

Development

Security & Best practices



Quarterly Active Contributors

187 active contributors in the last quarter - Project benefits from a large contributor base, ensuring continuous improvement and a vibrant development community



Quarterly Contributor Retention Rate

17% of contributors are contributing quarter over quarter - This project has strong contributor retention, reflecting an engaged and consistent contributor base.



Contributor Dependency

5 contributors account for 51%+ of contributions - This project has a moderate spread of contributions across several individuals.

Health Score Healthy

Share your project Health Score in your GitHub page.

[Generate badge](#)

About

A comprehensive open source platform designed to seamlessly manage PostgreSQL databases within Kubernetes environments.

First commit

February 2020

Software value ?

\$85M

Tags

No tags available

Links

[cnpg.io](#)

[/cloudnative-pg](#)

[@CloudNativePg](#)

Data sources ?



Git



GitHub

CloudNativePG - License

CloudNativePG has been originally created **EDB**

CloudNativePG is 100% **open source** and community-driven

All components are available under the [Apache 2 license](#) on Github

CloudNativePG - Adopters

- **Google Cloud**

<https://cloud.google.com/kubernetes-engine/docs/tutorials/stateful-workloads/cloudnativepg>

- **Microsoft Azure**

<https://learn.microsoft.com/en-us/azure/aks/postgresql-ha-overview>

- **IBM**

- **EDB**

EDB CloudNativePG Cluster, EDB CloudNativePG Global Cluster, EDB PostgreSQL AI Cloud Service (DBaaS)

Full list [here](#)

CloudNativePG - End users

- **EDB**
- **Hitachi**
- **Akamai Technologies**
- **Tembo**
- **CNDI**

Full list at https://cloudnative-pg.io/end_users

CloudNativePG - Beyond the operator

CloudNativePG project contains a plenty of interesting repositories:

- Helm charts
- CNPG-I
- Grafana dashboards
- PostGIS Containers
- Ciclops

CloudNativePG - Useful links

Website: <https://cloudnative-pg.io/>

CloudNativePG is on LinkedIn, GitHub, cncf.io/sandbox-projects/ , but also on

- **Slack** <https://cloud-native.slack.com/archives/C08MAUJ7NPM>
- **Twitter** <https://twitter.com/CloudNativePg>
- **Mastodon** <https://mastodon.social/@CloudNativePG>
- **Bluesky** <https://bsky.app/profile/cloudnativepg.bsky.social>

CloudNativePG - Resources

- CloudNativePG [website](#) and [GitHub](#) project
- CloudNativePG Recipes on Gabriele Bartolini's [blog](#)
- CloudNativePG maintainers and contributors are active on [Data On Kubernetes](#) community
- CNCF blog article "[Cloud Neutral Postgres Databases with Kubernetes and CloudNativePG](#)"
- Whitepaper with a benchmark of a PostgreSQL database in Kubernetes with CloudNativePG "[EDP Postgres AI on OpenShift cluster using CSI driver for Dell PowerFlex](#)"



**Join the
Community**



Let's start the workshop!

[https://github.com/gabriele-wolfox/
get-your-hands-dirty-with-cnpg](https://github.com/gabriele-wolfox/get-your-hands-dirty-with-cnpg)



Get in touch with me on

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- LinkedIn
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